

The money block of the model

The goal

- We want to see how the interest rate is determined in the model
- There is no interest rate equation, such as the Taylor rule
- Instead, the interest rate is given by the equilibrium on the money market
- Both money demand and money supply depend on the interest rate
- The interest will be that which sets money demand equal to money supply

Real money growth trend

- In the model, money supply is about base money, while money demand is about broad money, M2
- The difference between the two (in logarithm) is the money multiplier mm_t , which is given exogenously
- We assume that the multiplier is constant in the long run, which implies that the *trend growth rate* of base money and M2 are the same
- We call this trend growth simply *real money trend growth*, and denote it by $\Delta r\bar{m}_t$, given by the basic quantity equation

$$\Delta r\bar{m}_t = \Delta \bar{y}_t - \Delta v_t$$

i.e. the difference of potential growth and the change in velocity, both given exogenously

Money supply target

- The the long term nominal base money target growth rate is given by

$$\Delta \bar{m}b_t^s = \Delta r\bar{m}_t + \bar{\pi}_t + \bar{\varepsilon}_t \quad (1)$$

where $\bar{\pi}_t$ is the (implicit) inflation target, given exogenously

- This is basically the Friedman rule (or more precisely, Friedman's k -percent rule)
 - Nominal money growth should be based on long term fundamentals, irrespective of the cyclical position of the economy

Actual money supply

- But the central bank may chose to temporarily deviate from the long term target
- Reasons to do so can be
 - Inflation is too high
 - Demand is too low
 - The interest rate is different from the target (see later)
 - The government needs financing
- There is also persistency in deviation
- The model equations has only the last two components

$$\Delta mb_t^s = q_{11} \Delta mb_{t-1}^s + (1 - q_{11}) \left(\Delta \bar{mb}_t^s + q_5 (i_t - \bar{i}_t) + q_9 d\tilde{e}f_t \right) + \varepsilon_t^s \quad (2)$$

The target interest rate

- The interest rate target is given by a Taylor rule

$$\bar{i}_t = d_1 \bar{i}_{t-1} + (1 - d_1) \left(i_t^n + d_2 (\pi_{t+1} - \bar{\pi}_{t+1}) + d_3 \tilde{y}_t \right) \quad (3)$$

- The target rate goes up when
 - Inflation is higher than the inflation target
 - The output gap is positive
- The central bank tries to cool down the economy
- There is interest rate smoothing
- The long run value of the interest is equal to the neutral rate

$$i_t^n = \bar{r}_t + \bar{\pi}_{t+1} \quad (4)$$

Money demand

- Real money demand (for M2) is decomposed into the sum of a gap and a trend

$$rm2_t^d = r\bar{m}2_t^d + r\tilde{m}2_t^d \quad (5)$$

- Note that the *growth rate* of the trend is the same as for the money supply target

$$\Delta r\bar{m}2_t^d = \Delta r\bar{m}_t \quad (6)$$

- But when we calculate nominal money demand, we use the actual price level

$$m2_t^d = rm2_t^d + p_t \quad (7)$$

Real money demand gap

- Real money demand can temporarily deviate from its long run value due to
 - Transactional motives
 - Liquidity motives
- We proxy the first one by the output gap, the second one by the real interest rate gap, including the credit premium gap
- And again there is some persistence
- So the equation will be

$$r\tilde{m}2_t^d = q_{10} r\tilde{m}2_{t-1}^d + q_6 \tilde{y}_t - q_7 (\tilde{r}_t + pr\tilde{e}m_t) + \varepsilon_t^d \quad (8)$$

- We also have the real interest rate equations

$$r_t = \bar{r}_t + \tilde{r}_t \quad (9)$$

$$r_t = i_t - \pi_{t+1} \quad (10)$$

Variable counting

- The following variables are given "exogenously" (i.e. they have their own equations in other blocks of the model)

$$\Delta r\bar{m}_t, \quad mm_t, \quad \bar{r}_t, \quad \bar{\pi}_t, \quad \pi_t, \quad \tilde{y}_t, \quad \tilde{def}_t, \quad pr\tilde{e}m_t$$

- We have the following "endogenous variables" in the money block

$$\Delta \bar{m}b_t^s, \quad \Delta mb_t^s, \quad i_t, \quad \bar{i}_t, \quad i_t^n$$

$$\Delta r\bar{m}2_t^d, \quad rm2_t^d, \quad m2_t^d, \quad r\tilde{m}2_t^d, \quad r_t, \quad \tilde{r}_t$$

- That's 11 variables, and we have 10 equations

Equilibrium

- The missing equation is the equilibrium condition of the money market

$$m2_t^d = mb_t^s + mm_t \quad (11)$$

- All of the endogenous variables have their own equations, *except for the interest rate*
- *The interest rate is determined by the equilibrium condition*
- In the absence of an "own equation", the interest rate reaction to other variables, most importantly inflation, is not easy to examine
- But we will try, starting from the equilibrium condition, and then substituting the other equations

Solving for the interest rate I

- First take the equilibrium condition

$$m2_t^d = mb_t^s + mm_t \quad (12)$$

- Then use the definition of nominal money demand

$$rm2_t^d + p_t = mb_t^s + mm_t \quad (13)$$

- And now the actual money supply

$$\begin{aligned} rm2_t^d + p_t = & mm_t + mb_{t-1}^s + q_{11} \Delta mb_{t-1}^s / 4 \\ & + (1 - q_{11}) \left(\Delta \bar{mb}_t^s + q_5 (i_t - \bar{i}_t) + q_9 d\tilde{e}f_t \right) / 4 + \varepsilon_t^s / 4 \end{aligned} \quad (14)$$

Solving for the interest rate II

- Next, put in real money demand

$$r\bar{m}2_t^d + r\tilde{m}2_t^d + p_t = mm_t + mb_{t-1}^s + q_{11} \Delta mb_{t-1}^s / 4 + (1 - q_{11}) (\Delta \bar{m}b_t^s + q_5 (i_t - \bar{i}_t) + q_9 d\tilde{e}f_t) / 4 + \varepsilon_t^s / 4 \quad (15)$$

- Continue with real money demand gap

$$\begin{aligned} r\bar{m}2_t^d + q_{10} r\tilde{m}2_{t-1}^d + q_6 \tilde{y}_t - q_7 (\tilde{r}_t + pr\tilde{e}m_t) + \varepsilon_t^d + p_t &= \\ &= mm_t + mb_{t-1}^s \\ &+ q_{11} \Delta mb_{t-1}^s / 4 + (1 - q_{11}) (\Delta \bar{m}b_t^s + q_5 (i_t - \bar{i}_t) + q_9 d\tilde{e}f_t) / 4 + \varepsilon_t^s / 4 \end{aligned} \quad (16)$$

Solving for the interest rate III

- And then substitute out the real interest rate gap

$$\begin{aligned}
 r\bar{m}2_t^d + q_{10} r\tilde{m}2_{t-1}^d + q_6 \tilde{y}_t - q_7 (r_t - \bar{r}_t + pr\tilde{e}m_t) + \varepsilon_t^d + p_t &= \\
 &= mm_t + mb_{t-1}^s + q_{11} \Delta mb_{t-1}^s / 4 \\
 &+ (1 - q_{11}) (\Delta \bar{m}b_t^s + q_5 (i_t - \bar{i}_t) + q_9 d\tilde{e}f_t) / 4 + \varepsilon_t^s / 4
 \end{aligned} \tag{17}$$

- And finally put in the real interest rate definition

$$\begin{aligned}
 r\bar{m}2_t^d + q_{10} r\tilde{m}2_{t-1}^d + q_6 \tilde{y}_t - q_7 (i_t - \pi_{t+1} - \bar{r}_t + pr\tilde{e}m_t) + \varepsilon_t^d + p_t &= \\
 &= mm_t + mb_{t-1}^s + q_{11} \Delta mb_{t-1}^s / 4 \\
 &+ (1 - q_{11}) (\Delta \bar{m}b_t^s + q_5 (i_t - \bar{i}_t) + q_9 d\tilde{e}f_t) / 4 + \varepsilon_t^s / 4
 \end{aligned} \tag{18}$$

Solving for the interest rate IV

- Now start rearranging, first expand the products

$$\begin{aligned}
 r\bar{m}2_t^d + q_{10} r\tilde{m}2_{t-1}^d + q_6 \tilde{y}_t - q_7 i_t + q_7 (\bar{r}_t + \pi_{t+1} - pr\bar{e}m_t) + \varepsilon_t^d + p_t &= \\
 &= mm_t + mb_{t-1}^s + q_{11} \Delta mb_{t-1}^s/4 \\
 &+ (1 - q_{11}) \Delta \bar{m}b_t^s/4 + (1 - q_{11}) q_5 i_t/4 - (1 - q_{11}) q_5 \bar{l}_t/4 \\
 &+ (1 - q_{11}) q_9 d\tilde{e}f_t/4 + \varepsilon_t^s/4
 \end{aligned} \tag{19}$$

- Next, move the interest rate terms to the right hand side, and everything else to the left hand side

$$\begin{aligned}
 r\bar{m}2_t^d + q_{10} r\tilde{m}2_{t-1}^d + q_6 \tilde{y}_t + q_7 (\bar{r}_t + \pi_{t+1} - pr\bar{e}m_t) + \varepsilon_t^d + p_t \\
 - mm_t - mb_{t-1}^s - q_{11} \Delta mb_{t-1}^s/4 - (1 - q_{11}) \Delta \bar{m}b_t^s/4 \\
 + (1 - q_{11}) q_5 \bar{l}_t/4 - (1 - q_{11}) q_9 d\tilde{e}f_t/4 - \varepsilon_t^s/4 = \\
 = ((1 - q_{11}) q_5/4 + q_7) i_t
 \end{aligned}$$

Solving for the interest rate V

- Write this equation in a more readable format

$$\begin{aligned}
 & ((1 - q_{11}) q_5 / 4 + q_7) i_t = \\
 & + r \bar{m} 2_t^d \\
 & + q_{10} r \tilde{m} 2_{t-1}^d \\
 & + q_6 \tilde{y}_t \\
 & + q_7 (\bar{r}_t + \pi_{t+1} - pr \tilde{e} m_t) \\
 & + p_t \\
 & - m m_t \\
 & - m b_{t-1}^s \\
 & - q_{11} \Delta m b_{t-1}^s / 4 \\
 & - (1 - q_{11}) \Delta \bar{m} b_t^s / 4 \\
 & + (1 - q_{11}) q_5 \bar{l}_t / 4 \\
 & - (1 - q_{11}) q_9 d \tilde{e} f_t / 4 \\
 & - \varepsilon_t^s / 4 + \varepsilon_t^d
 \end{aligned} \tag{21}$$

Solving for the interest rate VI

- It is a bit disturbing to have levels on the right hand side, so let's do some rearrangement

$$\begin{aligned}
 r\bar{m}2_t^d - mm_t + p_t - mb_{t-1}^s &= \\
 r\bar{m}2_t^d - mm_t + p_t - p_{t-1} + p_{t-1} - mb_{t-1}^s &= \\
 r\bar{m}2_t^d - mm_t + \pi_t/4 - (mb_{t-1}^s - p_{t-1}) &= \\
 r\bar{m}2_t^d - mm_t + \pi_t/4 - rmb_{t-1}^s &= \\
 r\bar{m}2_{t-1}^d + \Delta r\bar{m}2_t^d/4 - mm_t + \pi_t/4 - rmb_{t-1}^s &= \\
 r\bar{m}2_{t-1}^d - rmb_{t-1}^s - mm_t + \Delta r\bar{m}_t/4 + \bar{\pi}_t/4 + \pi_t/4 - \bar{\pi}_t/4 &= \\
 r\bar{m}2_{t-1}^d - rmb_{t-1}^s - mm_t + \Delta \bar{m}b_t^s/4 - \bar{\varepsilon}_t/4 + \pi_t/4 - \bar{\pi}_t/4 &= \\
 r\bar{m}2_{t-1}^d - rmb_{t-1}^s - mm_{t-1} - (mm_t - mm_{t-1}) + \Delta \bar{m}b_t^s/4 - \bar{\varepsilon}_t/4 + \pi_t/4 - \bar{\pi}_t/4 &= \\
 -r\bar{m}2_{t-1}^d - \Delta mm_t + \Delta \bar{m}b_t^s/4 - \bar{\varepsilon}_t/4 + \pi_t/4 - \bar{\pi}_t/4 &
 \end{aligned}
 \tag{22}$$

Solving for the interest rate VII

- So we have

$$\begin{aligned}
 ((1 - q_{11}) q_5/4 + q_7) i_t = & \\
 & - r \tilde{m} 2_{t-1}^d \\
 & - \Delta m m_t \\
 & + \Delta \bar{m} b_t^s/4 \\
 & + \pi_t/4 - \bar{\pi}_t/4 \\
 & + q_{10} r \tilde{m} 2_{t-1}^d \\
 & + q_6 \tilde{y}_t \\
 & + q_7 (\bar{r}_t + \pi_{t+1} - p r \tilde{e} m_t) \\
 & - q_{11} \Delta m b_{t-1}^s/4 \\
 & - (1 - q_{11}) \Delta \bar{m} b_t^s/4 \\
 & + (1 - q_{11}) q_5 \bar{l}_t/4 \\
 & - (1 - q_{11}) q_9 d \tilde{e} f_t/4 \\
 & - \varepsilon_t^s/4 + \varepsilon_t^d - \bar{\varepsilon}_t/4
 \end{aligned} \tag{23}$$

Solving for the interest rate VIII

- And we can further combine terms to get

$$\begin{aligned}
 ((1 - q_{11}) q_5 / 4 + q_7) i_t = & \\
 & - \Delta m m_t \\
 & + \pi_t / 4 - \bar{\pi}_t / 4 \\
 & - (1 - q_{10}) r \tilde{m} 2_{t-1}^d \\
 & + q_6 \tilde{y}_t \\
 & + q_7 (\tilde{r}_t + \pi_{t+1} - p r \tilde{e} m_t) \\
 & - q_{11} \Delta m b_{t-1}^s / 4 \\
 & + q_{11} \Delta \bar{m} b_t^s / 4 \\
 & + (1 - q_{11}) q_5 \bar{l}_t / 4 \\
 & - (1 - q_{11}) q_9 d \tilde{e} f_t / 4 \\
 & - \varepsilon_t^s / 4 + \varepsilon_t^d - \bar{\varepsilon}_t / 4
 \end{aligned} \tag{24}$$

Solving for the interest rate IX

- Things are a bit simpler if we set $q_{11} = 0$, i.e. remove the smoothing from the money supply

$$\begin{aligned}
 (q_5/4 + q_7) i_t = & \\
 & - \Delta m m_t \\
 & + \pi_t/4 - \bar{\pi}_t/4 \\
 & - (1 - q_{10}) r \tilde{m} 2_{t-1}^d \\
 & + q_6 \tilde{y}_t \\
 & + q_7 (\bar{r}_t + \pi_{t+1} - pr\tilde{e}m_t) \\
 & + q_5 \bar{l}_t/4 \\
 & - q_9 d\tilde{e}f_t/4 \\
 & - \varepsilon_t^s/4 + \varepsilon_t^d - \bar{\varepsilon}_t/4
 \end{aligned} \tag{25}$$

Reaction of the interest rate I

- To see the immediate reaction of the interest rate to shocks, we can remove trends and predetermined variables

$$\begin{aligned}
 (q_5/4 + q_7) i_t = & \\
 & + \pi_t/4 - \bar{\pi}_t/4 \\
 & + q_6 \tilde{y}_t \\
 & + q_7 \pi_{t+1} \\
 & + q_5 \bar{l}_t/4 \\
 & - q_9 d\tilde{e}f_t/4 \\
 & - \varepsilon_t^s/4 + \varepsilon_t^d - \bar{\varepsilon}_t/4
 \end{aligned}
 \tag{26}$$

Reaction of the interest rate II

- We can go one step further, and substitute in the deficit to get

$$\tilde{def}_t = def_t - \bar{def}_t = def_t^{cyc} + \varepsilon_t^{def} = -r_2 \tilde{y}_t + \varepsilon_t^{def} \quad (27)$$

- Also, we can remove trends and predetermined variables from the target rate

$$\begin{aligned} \bar{i}_t &= d_1 \bar{i}_{t-1} + (1 - d_1) \left(i_t^n + d_2 (\pi_{t+1} - \bar{\pi}_{t+1}) + d_3 \tilde{y}_t \right) \\ &= (1 - d_1) d_2 \pi_{t+1} + (1 - d_1) d_3 \tilde{y}_t \end{aligned} \quad (28)$$

Reaction of the interest rate III

- So in the end we get

$$\begin{aligned}
 ((1 - q_{11}) q_5 + q_7) i_t = & \\
 & + \pi_t - \bar{\pi}_t \\
 & + (q_6 + (1 - q_{11}) q_9 r_2 + (1 - q_{11}) q_5 (1 - d_1) d_3) \tilde{y}_t \quad (29) \\
 & + (q_7 + (1 - q_{11}) q_5 (1 - d_1) d_2) \pi_{t+1} \\
 & - \varepsilon_t^s + \varepsilon_t^d - (1 - q_{11}) q_9 \varepsilon_t^{def}
 \end{aligned}$$

- And this is almost like a Taylor rule

Special case: classical money targeting I

- Supporters of classical money targeting suggest that the central bank should never deviate from the long term money growth target
 - In the language of the model, this means $q_{11} = 0$, $q_5 = 0$ and $q_9 = 0$
 - Also, there are no supply shocks, i.e. $\varepsilon_t^s = 0$, and money supply is always on target
- They also believe that there is a stable relationship between interest rates and money demand
 - In the language of the model, this means $q_{10} = 0$, $q_6 = 0$ and q_7 not very close to 0

Special case: classical money targeting II

- So the interest rate equation becomes

$$\begin{aligned}
 q_7 i_t = & \\
 & + r \bar{m}_t \\
 & + q_7 \pi_{t+1} \\
 & + q_7 \bar{r}_t \\
 & + p_t \\
 & - m_{t-1}^s \\
 & - \Delta r \bar{m}_t \\
 & - \bar{\pi}_t \\
 & + \varepsilon_t^d
 \end{aligned}
 \tag{30}$$

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